

LHE
Environmental
Hydraulics
Laboratory

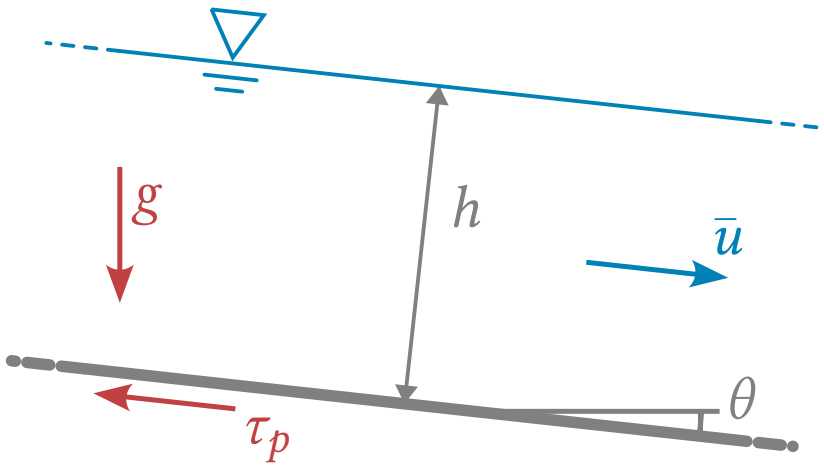
Hyporheic flow and velocity distribution

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Flow resistance



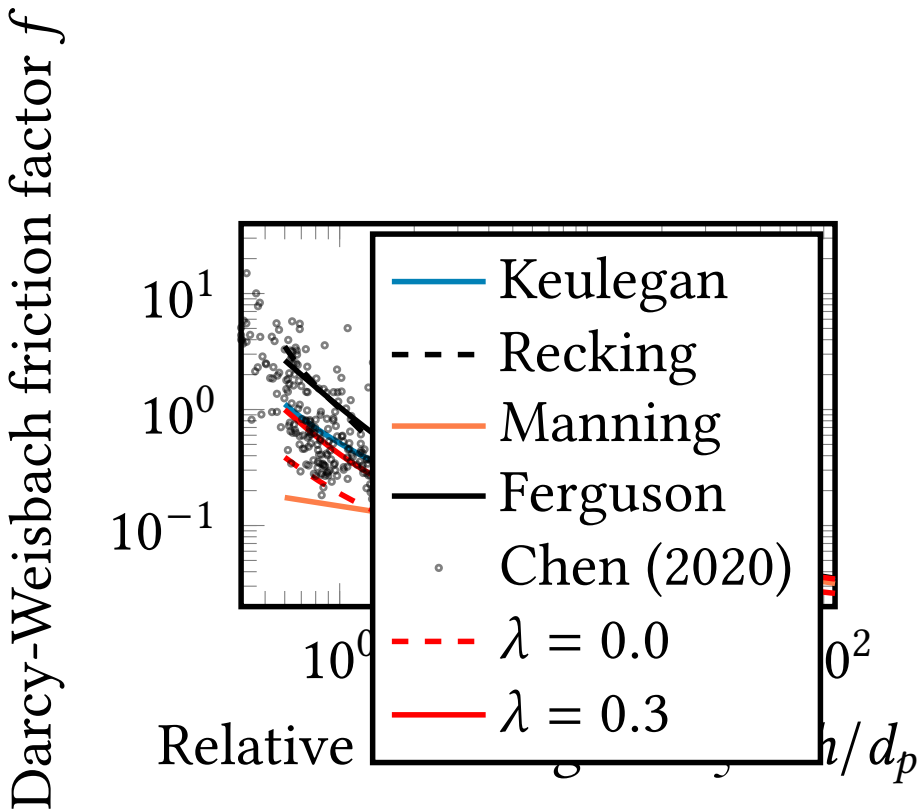
According to the (Gauckler-)Manning(-Strickler) formula, the bed's shear resistance is :

$$\tau_p = \frac{\varrho g}{K^2} \frac{\bar{u}^2}{h^{1/3}}$$

- ϱ : the fluid's density,
- g : the gravitational acceleration,
- K : a calibrated coefficient (depends on roughness),
- $\bar{u}$: the mean flow speed and
- h : the flow depth.

Gauckler, "Du mouvement de l'eau dans les conduites."

(Gauckler-)Manning(-Strickler)'s validity domain



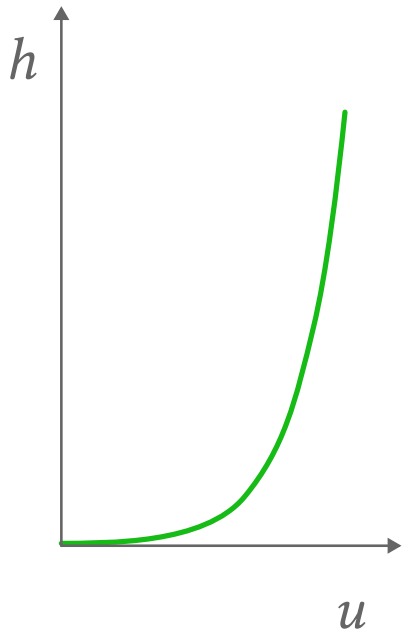
Recking et al., "Feedback between bed load transport and flow resistance in gravel and cobble bed rivers"

Ferguson and Recking, "Flow Resistance and Hydraulic Geometry in Gravel-And Boulder-Bed Rivers"

Chen et al., "Rough Correlations: Meta-Analysis of Roughness Measures in Gravel Bed Rivers"

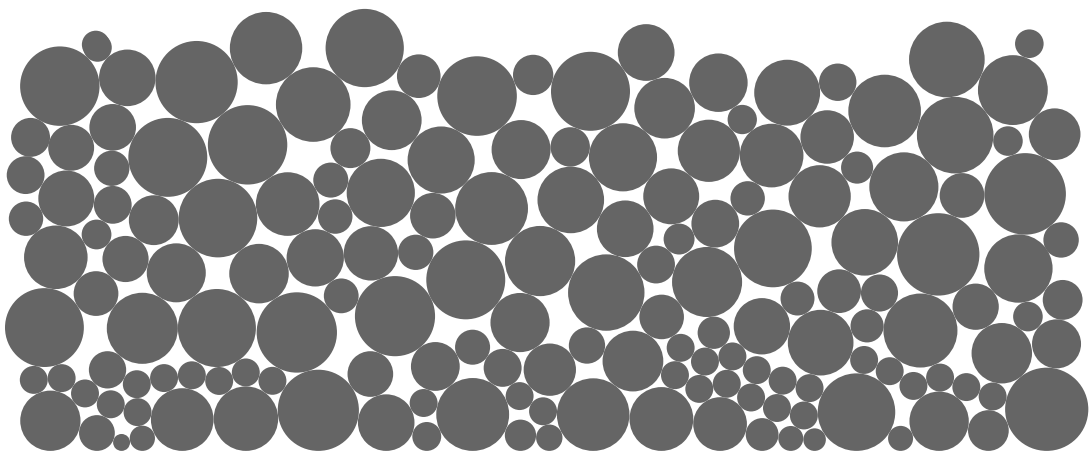
Velocity profile in steep gravel bed rivers

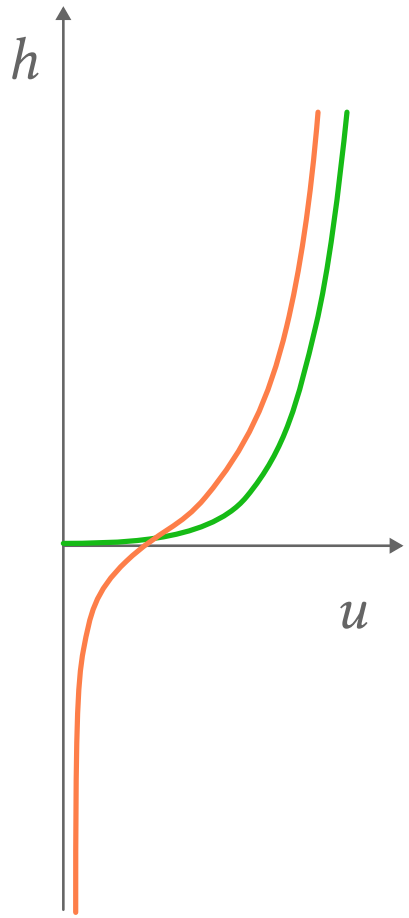




Velocity profile in steep gravel bed rivers

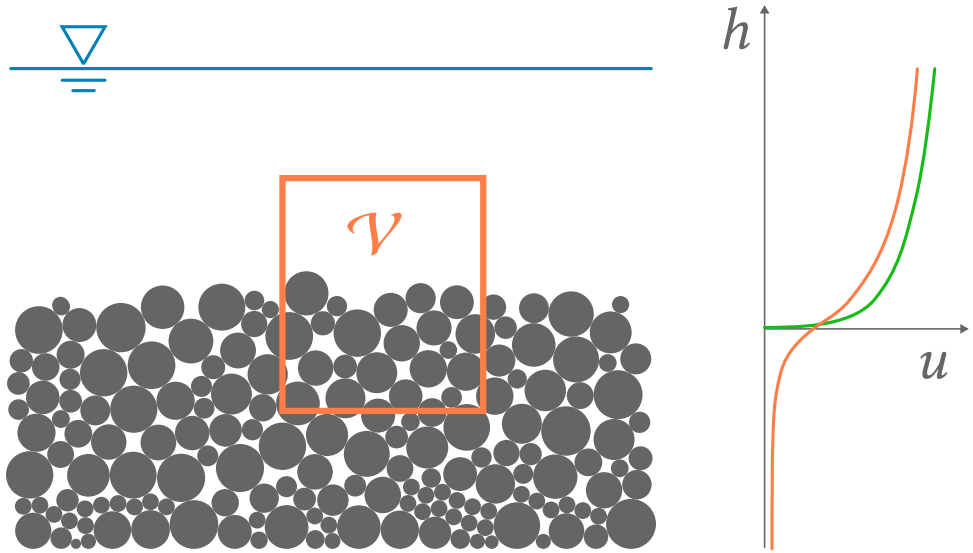






How can hyporheic flow be included ?

Representative Elementary Volume



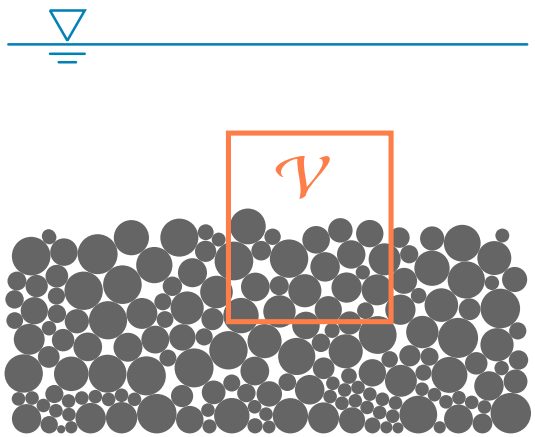
$$\varrho \frac{\partial u}{\partial t} + \varrho (u \cdot \nabla) u = \varrho g - \nabla p + \nabla \cdot (\tau_v + \tau_t) \,, \qquad u \leftarrow \frac{1}{V} \int_{\mathcal{V}} u \, \mathrm{d}V$$

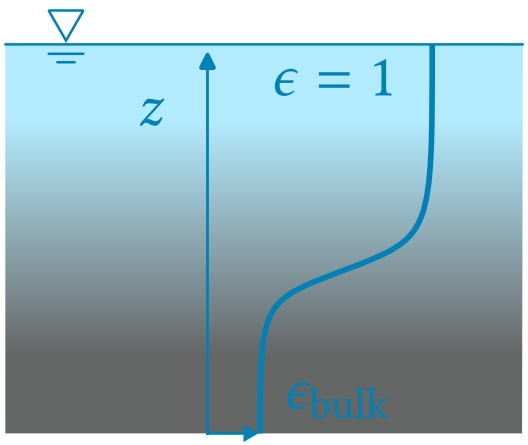
$$\varrho \frac{\partial u}{\partial t} + \varrho (u \cdot \nabla) u = \varrho g - \nabla p + \nabla \cdot (\tau_v + \tau_t + \tau_d) + \epsilon f - \mu \nabla u \nabla \epsilon$$

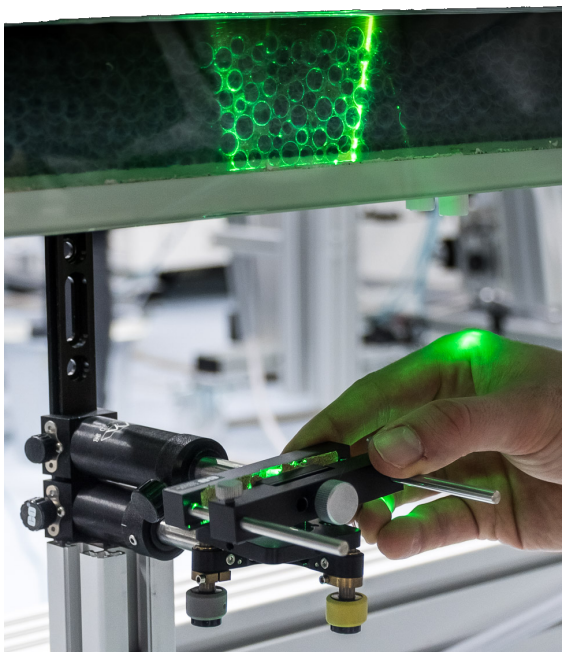
Nikora et al., “Spatially Averaged Open-Channel Flow over Rough Bed”

Steady-state uniform flow equation

A single ODE







$$0 = \epsilon \varrho g i + \frac{\mathrm{d}(\tau_v + \tau_t + \tau_d)}{\mathrm{d}z} + \epsilon f - \mu \frac{\mathrm{d}u}{\mathrm{d}z} \frac{\mathrm{d}\epsilon}{\mathrm{d}z}$$

Rousseau and Ancey, “An experimental investigation of turbulent free-surface flows over a steep permeable bed”

Differential equation to solve

$$0 = \epsilon\varrho g i + (\tau_v + \tau_t + \tau_d)' + \epsilon f - \mu u' \epsilon'$$

With closures for τ_t, τ_d and f:

$$\begin{cases} \tau_t = \varrho\epsilon\kappa^2 Z_{LS}^2 \left(1 - e^{-\sqrt{\frac{Z_{LS} u}{\nu A_*^2}}}\right)^2 u'^2, \\ \tau_d = \varrho\epsilon\lambda d_p \frac{1-\epsilon}{1-\epsilon_b} u u', \\ f = -\varrho\nu A_E \left(\frac{1-\epsilon}{\epsilon d_p}\right)^2 u - \varrho B_E \frac{1-\epsilon}{d_p} u^2. \end{cases}$$

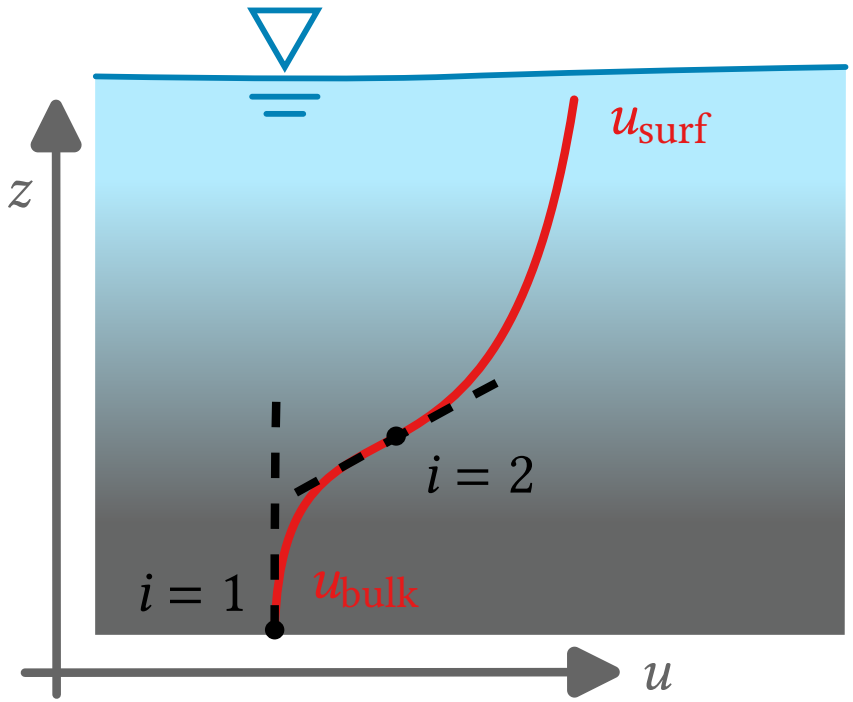
$$\Rightarrow 0 = \underbrace{c_0 + c_1 u' + c_2 u'^2 + c_3 u'^3 + \overbrace{c_4 u' u''}^{\text{Affine of } u''}}_{\text{Trinomial of } u'} + c_5 u''$$

$$c_i = c_i(z, u)$$

Approximate analytical solution

Asymptotic expansions

$$u_i = u_i^{(0)} + \alpha_i u_i^{(1)} + O(\alpha_i^2)$$



Conclusion

- Hyporheic flow reduces the derivative of the velocity profile.
- Volume averaging tranforms the biphasic media into a continuous porous media.
  - $\longrightarrow$ Yields a single ODE valid over the entire depth.

$\hookrightarrow$ Current step : deriving the matched asymptotic expansions of the velocity profile's ODE.

References

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